

**MTH 3340 Abstract Algebra, Section 001, 3 Credits**  
**Spring 2026**

**Professor:** Zach Bryhtan

**Office:** Library room 210 (L210)

**Office Hours:** By appointment or

**Mondays:** 10:00 AM – 11:50 AM

**Tuesdays, Thursday:** 9:00 PM – 9:50 PM

**Email:** [bryhtan@msoe.edu](mailto:bryhtan@msoe.edu)

**Course meetings:** 12:00 PM – 12:50 PM MWF in the Walter Schroeder Library room 307 (L307)

**Course Materials/Required Textbook:**

**Title:** *Abstract Algebra: Theory and Applications, Annual Edition 2022*

**Author:** Thomas W. Judson

**Link:** <http://abstract.pugetsound.edu/>

**Note:** The textbook we will be using is an open source textbook. You may use the link above to view the book or download a PDF copy.

**Course Description:** This course covers methods of mathematical reasoning and proof. It also covers elementary number theory including divisibility, primes, congruence and modular arithmetic, and Euler's formula. It includes introduction to group theory including subgroups, symmetry and permutation groups, homomorphism and isomorphism, direct products, structure of cyclic and abelian groups, and group actions. Applications to cryptography are addressed. Other possible topics include rings and fields, solvability of polynomials, impossibility of compass and straight edge constructions, or primality testing.

**Prereq:** MTH 2340 or MTH 2140 or MTH 2310

**Course Learning Outcomes:**

*Upon successful completion of this course, the student will be able to:*

- Prove elementary facts in number theory
- Solve linear congruence equations
- Determine if a given set with a binary operation is a group
- Determine if a given subset of a group is a subgroup
- Determine the possible orders of elements and subgroups of a given group
- Determine the cycle structure and orders of elements in a permutation group
- Determine if two groups are isomorphic; recognize standard classes of groups
- Determine the kernel and image of a group homomorphism; the cosets of a normal subgroup and the associated quotient group
- Describe the structure of a finite abelian group as a direct sum
- Use the RSA algorithm to encrypt and decrypt messages

**Grading:** Your final grade will be determined by your attendance and performance on take-home quizzes, midterm exams, and the final exam. These categories are weighted according to the following table.

| Quizzes | Midterm 1 | Midterm 2 | Midterm 3 | Final Exam |
|---------|-----------|-----------|-----------|------------|
| 15%     | 20%       | 20%       | 20%       | 25%        |

**Homework:** Homework will be assigned during class, *but it will not be collected*. It is absolutely necessary for you to keep up with the homework assigned in order to succeed in the course. You can utilize the tutoring center or come to my office if you have any questions.

**Quizzes:** There will be a short take-home quiz assigned every week except for the first week, and weeks of exams. Quizzes will be given on Monday in class and due at the start of Friday's class of the same week. Each quiz will present problems similar to the problems from the previous week's homework problems.

*No quiz scores will be dropped.*

**Midterm Exams:** There will be three 50-minute midterm exams. Each exam will be given during the regular class period.

Exam 1: Monday, February 16<sup>th</sup>

Exam 2: Monday, March 23<sup>rd</sup>

Exam 3: Monday, April 20<sup>th</sup>

**Final Exam:** The final exam is a comprehensive exam scheduled during finals week (May 11<sup>th</sup> – May 15<sup>th</sup>).  
*Time and day of the exam TBA.*

**Attendance Policy:** Attending class regularly and engaging in discussions during lectures are crucial for success in this course. I will take attendance and, together with class participation, may be a factor in determining your final grade.

**Late Work Policy:** Take-home quizzes may be turned in no later than 1 week for partial credit. Late quizzes are only eligible for a maximum grade of 70%.

**Make-Up Policy:**

**Quizzes:** There will be no make-up quizzes allowed. The only exception to this policy is if you are absent due to MSOE related event (athletic meet, student organization convention, etc.) or have a documented medical excuse.

**Exams:** A make-up for exams may be granted in extenuating circumstances. Please contact me as soon as possible if you believe you may have to miss an exam.

**Grading Scale:**

| F       | D        | CD       | C        | BC       | B        | AB       | A       |
|---------|----------|----------|----------|----------|----------|----------|---------|
| [0, 70) | [70, 74) | [74, 77) | [77, 81) | [81, 85) | [85, 89) | [89, 93) | [93, ∞) |

**Institutional Policies:**

- **Nondiscrimination statement:** Milwaukee School of Engineering admits students of any race, color, national and ethnic origin to all the rights, privileges, programs and activities generally accorded, or made available, to students at the university. It does not discriminate on the basis of race, color, national and ethnic origin, religion, age, gender, sexual orientation, marital status or disability in administration of its educational policies, admission policies, scholarship and loan programs, and athletic and other institutionally administered programs. MSOE also maintains its long-standing policy as an Equal Opportunity/Affirmative Action Employer of male and female personnel for its faculty and administrative staff.
- **Student Accessibility Services (SAS):** For students with documented disabilities, chronic medical conditions or mental health concerns; MSOE provides services to make reasonable accommodations available. If you are a student who requires or anticipates the need for accommodations, please contact Student Accessibility Services Office at 414-277-7281, by email at [moureau@msoe.edu](mailto:moureau@msoe.edu), or in person at K250 to discuss appropriate accommodations and eligibility requirements.
- **Policy on Student Integrity** (Academic Integrity)
- **[Sexual Misconduct Policy](#)**
- **Research with Human Participants** (Responsible Conduct of Research)  
[Link to the 2025-26 undergraduate catalog policy page](#)